Software Test Plan (STP)
Student Attendance Management System (SAMS)

Project: Student Attendance Management System (SAMS)
Version: 1.1
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1. Introduction

Purpose

This document defines the test plan for the Student Attendance Management System (SAMS) v1.0. It outlines the objectives, scope, strategy, resources, schedule, and responsibilities for testing the system as specified in the Software Requirements Specification (SRS), Team 14.

Scope

Testing covers all functional areas of SAMS described in the SRS: user authentication and role-based access control (Admin, Teacher, Student), course and class management, student enrollment management, attendance marking and recording (including corrections), attendance reports and analytics, and notifications and leave management. Testing also covers the associated non-functional requirements (performance, reliability, usability) and security requirements (input validation, unauthorized access/modification prevention, role-based data access, and session expiry). Biometric or hardware-based attendance capture, payroll integration, and mobile push-notification infrastructure are explicitly out of scope, as the system is a web-based application accessed through a standard browser.

References

- Software Requirements Specification (SRS) – Student Attendance Management System (SAMS), Team 14, v1.0, 04-09-2026
- UML Use-Case Diagrams (UML1: Core Admin, Teacher, and Student use cases; UML2: Reporting and notification subsystem use cases)
- Requirements Traceability Matrix (RTM), SRS Section 8

Definitions

Student, Teacher, Admin, Attendance Record, Session/Period, Attendance Percentage, Defaulter, Correction Window, Roster, FR (Functional Requirement), NFR (Non-Functional Requirement), UML (Unified Modeling Language), RTM (Requirements Traceability Matrix) — as defined in SRS Section 1.4.

2. Test Items

The following modules, as defined in SRS Sections 2 and 8, are subject to test:

- AuthModule – login, authentication, role-based access control, session expiry
- UserModule – Admin management of Teacher and Student accounts

- CourseModule – course, section, and schedule creation; Teacher assignment
- EnrollmentModule – student enrollment, transfer, and removal with history preservation
- AttendanceModule – attendance marking, metadata logging, correction window edits
- ReportModule – attendance percentage calculation, defaulter list generation, report export
- NotificationModule – low-attendance alerts, leave request submission and approval
- UI Module – role-specific dashboards (Admin, Teacher, Student) and usability

3. Features to be Tested

Features mapped to SRS requirement IDs:

- SAMS-F-001 – User login with unique ID and password
- SAMS-F-002 – Role-based access control across Admin, Teacher, Student screens
- SAMS-F-003 – Admin creation, editing, and deactivation of Teacher/Student accounts
- SAMS-F-004 – Admin creation and management of courses, sections, and schedules
- SAMS-F-005 – Admin assignment of Teachers to courses/sections
- SAMS-F-006 – Admin enrollment of students and roster maintenance
- SAMS-F-007 – Student transfer/removal with preserved historical attendance
- SAMS-F-008 – Teacher marking of attendance (Present/Absent/Late) per session
- SAMS-F-009 – Recording of date, session/period, and marking Teacher identity
- SAMS-F-010 – Teacher edit of attendance within the correction window
- SAMS-F-011 – Attendance percentage calculation per course and overall
- SAMS-F-012 – Defaulter list generation against a configurable threshold
- SAMS-F-013 – Attendance report export (CSV/PDF) by course, section, date range
- SAMS-F-014 – Low-attendance alert notification to Students
- SAMS-F-015 – Student leave request submission and Teacher approval/rejection
- SAMS-NF-001 – Attendance submission-to-confirmation latency ≤ 3s (60 students, 95% of runs)
- SAMS-NF-002 – No data loss/duplication on network interruption or session timeout during submission
- SAMS-NF-003 – First-time Teacher completes marking a 40-student class within 2 minutes
- SAMS-NF-004 – 100 concurrent users with average response time ≤ 2s, no failed requests
- SAMS-NF-005 – Percentage/defaulter report generation ≤ 5s for 200 students (95% of runs)
- SAMS-NF-006 – 8+ hour continuous run without crash or response-time degradation
- SAMS-SR-001 – Validation of user-provided input (credentials, attendance data)
- SAMS-SR-002 – Prevention of unauthorized attendance record modification
- SAMS-SR-003 – Restriction of administrative/configuration operations to Admin accounts
- SAMS-SR-004 – Protection of stored student data from unauthorized access/disclosure
- SAMS-SR-005 – Automatic session expiry after 30 minutes of inactivity

4. Features Not to be Tested

- Biometric or hardware-based attendance capture (e.g., fingerprint or RFID devices) (explicitly out of scope per SRS Section 1.2)

- Payroll integration and mobile push-notification infrastructure (explicitly out of scope per SRS Section 1.2)
- Browser, operating-system, and third-party web-framework/database internals (assumed correct per vendor/library documentation)
- Underlying network/HTTPS transport infrastructure beyond the application's own communication layer (SRS Section 3.4)

5. Test Approach / Strategy

Levels

- Unit tests (module-level: AuthModule, AttendanceModule, ReportModule, NotificationModule, etc.)
- Integration tests (e.g., EnrollmentModule ↔ AttendanceModule, AttendanceModule ↔ ReportModule, NotificationModule ↔ AttendanceModule)
- System tests (end-to-end role workflows: login → course/roster setup → attendance marking → reporting/notification)
- Acceptance tests (per acceptance criteria defined in SRS Sections 4–5)

Types

- Functional testing (authentication, course/enrollment management, attendance marking/correction, reporting, notifications/leave)
- Regression testing (after each build/fix cycle)
- Performance testing (submission latency, concurrent-user load, report generation time)
- Usability testing (first-time Teacher marking workflow, role-specific dashboard clarity)
- Reliability testing (8-hour soak test, network-interruption/session-timeout handling)
- Security testing (input validation, unauthorized access/modification attempts, session expiry)

Entry Criteria

Stable build delivered; test environment set up (web application server, database, and notification/email service available); test data (dummy Admin/Teacher/Student accounts, sample courses, sections, and rosters) available; SRS and RTM baselined.

Exit Criteria

100% of planned test cases executed; 0 critical/high-severity defects open; all acceptance criteria in SRS Sections 4–6 satisfied; RTM shows full requirement coverage.

5.1 Security Validation

- Validate rejection of invalid, excessively long, or unsupported input in login and attendance data fields (SAMS-SR-001)
- Verify unauthorized attendance-modification attempts are rejected and the original record is preserved (SAMS-SR-002)
- Verify administrative and configuration operations are restricted to authenticated Admin accounts (SAMS-SR-003)
- Verify each role can access only the data permitted for it — Students their own records, Teachers their assigned sections, Admin all data (SAMS-SR-004)

- Verify authenticated sessions expire after 30 minutes of inactivity and require re-authentication (SAMS-SR-005)
- Attempt authentication bypass and boundary/negative-value testing on stored attendance and configuration data

6. Test Environment

Hardware

Desktop, laptop, or tablet device with a display supporting a minimum resolution of 1024×768 pixels, standard keyboard and mouse/touch input, and a stable internet or institutional network connection, per SRS Section 2.4.

Software

SAMS web application v1.0; a modern, supported web browser; the selected web server and database backend; the notification/email service used for low-attendance alerts and leave-request status updates.

Tools

- Manual and scripted test execution (browser-driven test scripts)
- Load-testing / concurrency-simulation tool (performance testing, SAMS-NF-004)
- Response-time / latency logging utility (performance testing)
- Defect tracking tool (e.g., Jira or equivalent)

Test Data

Dummy Admin, Teacher, and Student credentials; sample courses, sections, schedules, and rosters (including a 40-student and a 60-student section, and a 200-student section for report load testing); a range of valid/invalid/boundary attendance and configuration input values.

7. Test Schedule

Milestones (indicative):

Milestone	Target Date
Test case design	24-Sep-2026
Environment setup	26-Sep-2026
Test execution start	29-Sep-2026
Test execution end	15-Oct-2026
Acceptance / final review	17-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (mapped to SRS requirement IDs, per Sections 6.1–6.3 of the SRS)
- Test Scripts (manual procedures / any automated scripts)
- Test Data sets

- Test Execution Logs
- Defect Reports
- Requirements Traceability Matrix (updated with pass/fail status)
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Priyanka M Reddy	Prepare test plan, coordinate execution
Test Engineer	Priyanka N	Design & execute test cases, log defects
Developer	Priyanka, Priyanka Sajjan	Support defect fixes and triage
Course Coordinator	Asha Kumari	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds from the development team
Concurrent-load environment unavailable or unrepresentative of institutional scale	Use a load-testing/concurrency-simulation tool against a staging environment sized to SRS Section 5
Ambiguity in timing-based NFRs (e.g., submission latency) across network conditions	Run performance tests on a standard reference environment and document network conditions
Limited team availability for full regression coverage	Prioritize high-priority requirements (SAMS-F-001/002/003/004/006/008/009/011/012) first
Data loss/duplication defects during interrupted submissions are hard to reproduce	Use scripted network-interruption simulation and inspect stored records after each run

11. Assumptions & Dependencies

- The selected web framework, database, browser, and notification/email service behave as documented
- Test data (dummy Admin/Teacher/Student credentials, sample courses, sections, and rosters) will be available before execution
- No biometric/hardware attendance-capture, payroll, or mobile push-notification testing is required, consistent with SRS scope
- A stable, feature-complete build will be delivered ahead of each test cycle
- An environment capable of simulating at least 100 concurrent user sessions will be available for load testing (SAMS-NF-004)

12. Suspension & Resumption Criteria

Suspend testing if:

- The build is too unstable and blocks more than 30% of planned test cases
- The test environment (application server, database, or notification service) is unavailable for more than 4 hours

Resume testing if:

- Blocking defects are resolved
- The environment is stabilized and re-verified

13. Test Case Management & Traceability

The Requirements Traceability Matrix (RTM), as defined in SRS Section 8, maps every SRS requirement to its corresponding test case(s). Representative mappings:

Req ID	Requirement	Test Case(s)
SAMS-F-001/002	Login / role-based access control	TC-Auth-01, TC-Auth-02
SAMS-F-003	Admin account creation and deactivation	TC-Auth-03
SAMS-F-004/005	Course/section management, Teacher assignment	TC-Course-01, TC-Course-02
SAMS-F-006/007	Enrollment, transfer/removal with history	TC-Enroll-01, TC-Enroll-02
SAMS-F-008..010	Attendance marking, metadata, correction window	TC-Mark-01, TC-Mark-02, TC-Mark-03
SAMS-F-011..013	Percentage calculation, defaulter list, export	TC-Report-01, TC-Report-02, TC-Report-03
SAMS-F-014/015	Low-attendance alerts, leave request/approval	TC-Notify-01, TC-Notify-02
SAMS-NF-001..006	Performance, reliability, usability	TC-Perf-Mark-01, TC-Rel-Mark-01, TC-UX-Mark-01, TC-Perf-Load-01, TC-Perf-Report-01, TC-Rel-Soak-01
SAMS-SR-001..005	Input validation / access control / session expiry	TC-Sec-01..05

The full RTM (all 26 requirement IDs) is maintained in the SRS, Section 8, and shall be updated with Pass/Fail/Not-Run status throughout execution.

14. Test Metrics & Reporting

Metrics collected

- % of test cases executed
- % passed / failed
- Defect density and defect aging
- Requirement coverage (from the RTM)
- Submission latency, concurrent-user response time, and report-generation time against SAMS-NF-001/004/005

Reports

- Daily execution status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead		
Dev Lead		
Course Coordinator		